

Triage of the U-series verification script: five discrepancies classified, near-gap endpoint protection downgraded

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The autonomous session's consolidated verification script ran 52/57 on first execution. Five asserts failed. This note classifies each as **TYPO** / **ESTIMATE-DISCREPANCY** / **REAL-ERROR** / **UPHELD-BLOCKER**, records the corrected (truth-asserting) script (v0.2.0, 57/57), and propagates the one consequential finding to the U1 promotion proposal. No tier action is claimed; the U1 proposal is **DOWNGRADED** to blocked-pending-repair.

2. The five classifications

1. **m_w_window_constant** — **TYPO (math sound)**. The U2/ha0 note quoted $m_w = 0.005 + 48.6(43/1620)^2 = 0.0392414$; the true value is 0.0392407 (sixth-decimal typo). The script computed 0.0392407 correctly. Note corrected; no consequence.
2. **Mprime_equals_minus_J0** — **REAL factor-2 error**. U7/U10 asserted the exact identity $M'(\hat{r}) = -J(0)$. The finite-difference value is $-M' = 0.1452$ against $J(0) = 0.2896$: the true identity is $M'(\hat{r}) = -\frac{1}{2}J(0)$ (ratio 1.995). The qualitative content ($M' < 0$, so intensity dressing softens the kernel) is unaffected; the stated *exact* identity is off by $\times 2$ and is corrected in the script and flagged in U7/U10.
3. **U7_dressed_ratio (endpoint)** — **ESTIMATE discrepancy**. U7 quoted the dressed endpoint ratio ~ 1.34 ; the true value is 1.13 (downstream of the factor-2 normalisation of item 2). The endpoint remains > 1 (the **U7_endpoint_positive** assert passed): the intensity-dressed sunset bound still sits below the composed margin, but the quoted figure was optimistic.
4. **convention_remainder_endpoint** — **REAL ($\times 60$ overclaim)**. U11 claimed the near-gap convention remainder at $I = 2 \times 10^{-3}$ is $\sim 2.7 \times 10^{-5}$. The self-consistent LAM solve gives 1.65×10^{-3} — about $60\times$ larger. The remainder is NOT negligible at the endpoint.
5. **remainder vs protection** — **UPHELD BLOCKER (R-U10-3)**. Consequent to item 4: the near-gap protection ratio is $\text{gain} \cdot I / \text{remainder} = 3.5 \times 10^{-3} / 1.65 \times 10^{-3} \approx 2.1$, NOT the $\times 100 / \times 130$ claimed by U11. The near-gap small-amplitude endpoint is protected by only $\times 2$.

3. Consequence for the U1 T6 proposal

The U9 self-audit had UPHELD the objection G-U1-SMALLT (the near-gap small-amplitude regime was unverified in the U1 uniform-floor composition). U10/U11 claimed to repair it with a $\times 130$ margin; the triage shows the true endpoint margin is $\times 2$. Therefore:

U1 Reading-H T6-CONDITIONAL proposal: BLOCKED on R-U10-3
(near-gap endpoint, margin $\times 2$, not closed comfortably).

This is not a falsification ($\times 2 > 1$: the protection holds), but it is below the comfort threshold for a tier promotion. The repair options are: (i) sharpen the near-gap envelope (the $\times 2$ uses the crude sup-kernel; a per-transfer kernel should widen it); (ii) restrict the amended class's minimum amplitude; (iii) accept a $\times 2$ -margin explicitly-conditional T6 with the thin endpoint flagged. The choice is the operator's.

4. Numerical verification

Script `codes/vacuum/t6_mainline_useries_checks.py` (v0.2.0; 57/57 asserts PASS after the triage corrections; artefact `claims/B1-RH-ENUM/runs/260606-useries-checks/result.json`). The five corrected asserts now assert the TRUE values (item 1 typo; item 2 $M' = -J(0)/2$; item 3 ratio 1.13; item 4 remainder 1.65×10^{-3} ; item 5 protection $\times 2.1$), so the script certifies the corrected arithmetic rather than the over-optimistic note claims.

Quantitative sanity check (mandatory): the factor-2 of item 2 and the $\times 60$ of item 4 are independent (M' is a derivative identity; the remainder is a self-consistency gap), yet both reduce the SAME endpoint margin, which is why the protection collapses from the claimed $\times 130$ to $\times 2$; $3.5 \times 10^{-3}/1.65 \times 10^{-3} = 2.12$ reproduces the boxed figure.

5. Devil's-advocate (three objections; CLAUDE.md 6.3)

- (α) “Asserting the true values just hides the note errors behind a green script.” DISMISSED: the corrections are recorded HERE with the original claims and the deltas; the script comments carry the TRIAGE tag at each corrected assert; the affected notes (U7/U10/U11) are flagged for operator review. Green now means “arithmetic verified”, not “note claims vindicated”.
- (β) “ $\times 2$ may itself be sup-kernel-crude; the true margin could be fine.” VALID-with-mitigation: precisely why the repair option (i) is registered. But a tier promotion cannot rest on an un-sharpened $\times 2$; the proposal is blocked until the sharper kernel is computed, not assumed.
- (γ) “The factor-2 identity error might propagate further than item 3.” VALID-with-mitigation: items 2 and 3 are the only M' -dependent asserts; the composed-margin and floor results do not use M' (they use $J(|t|)$ directly). A full M' -usage grep is registered as a polish item.

6. Result footer

Result ID:	B1-RH-ENUM / U-series triage (gate R-U10-3)
Precise statement:	5 script FAILs classified: 1 TYPO (m_w), 1 REAL factor-2 ($M' = -J(0)/2$ not $-J(0)$), 1 ESTIMATE (U7 ratio 1.13 not 1.34), 1 REAL x60 overclaim

(remainder 1.65e-3 not 2.7e-5), 1 UPHELD BLOCKER
(near-gap endpoint protection x2 not x100/x130).
Script corrected to assert truth: v0.2.0, 57/57.

Scope: Autonomous-session U-series only; corrections are
arithmetic, not note-body rewrites (operator
review pending on U7/U10/U11).

Dependencies: U9 (G-U1-SMALLT), U10/U11 (near-gap), U7 (sunset),
U2 (window), AddC (J machinery).

Evidence grade: AUDIT + EXECUTED (57/57 corrected)

Reproduction command: python codes/vacuum/t6_mainline_useries_checks.py

Expected output: claims 57/57 PASS; result.json under
claims/B1-RH-ENUM/runs/260606-useries-checks/

Falsification gate: Any of the five corrected values failing on
re-run; a sharpened near-gap kernel giving margin
< 1 (would falsify the protection, not just block).

Tier before / after: B1 unchanged at T5; the U1 T6-CONDITIONAL
PROPOSAL is downgraded to BLOCKED-pending-repair.

No-overclaim statement: The near-gap protection HOLDS ($x_2 > 1$) but is too
thin for promotion; this note BLOCKS the U1
proposal, it does not close or falsify anything.
No tier action on B1/B2/B5; Reading-H stays T5.

Next required action: Operator: choose the R-U10-3 repair (sharper
kernel / min-amplitude restriction / explicit
 x_2 -conditional T6). Polish: M'-usage grep;
U7/U10/U11 note-body corrections.